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PlantSnap

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1. Summary

PlantSnap is an Ai-powered Android application that makes it possible for anyone to identify plants instantly. It serves two audiences: curious everyday users who want to learn about plants around them, and safety concerned outdoor enthusiasts and survivalists who need reliable information if a plant is edible, toxic or medicinal.

The application combines two AI systems, PlantNets computer vision model for species identification and a LLM for contextual plant information.

PlantSnap is built offline first: all core identification and history features work without an internet connection. Optional Supabase-backed cloud sync and user accounts allow cross-device history and profile features for signed-in users.

2. Problem Statement

2.1 The problem we are solving

Most people encounter plants they cannot identify every day, at home, hiking trails, in parks or growing wild near their homes.

- Identifying a plant manually requires expertise and field guides
- In survival or foraging scenarios, misidentifying a toxic plant can be life threatening
- Plant care advice is fragmented across dozens of websites with inconsistent quality
- Existing solutions are either too complex or too superficial

2.2 What has changed ?

Three technological developments make PlantSnap viable in 2026.

- Computer vision for plant identification is accurate and consumer grade
- LLMs can generate reliable, structured plant care and safety information
- Androids CameraX API makes came integration straightforward

3. Vision

3.1 Mission

To make plant knowledge viable for everyone by combining the accuracy of computer vision with depth of generative AI, delivered through an application that makes it easy and enjoyable for everyone.

3.2 Core values

Value	What it means
Accuracy	Identification results must be trustworthy
Safety	Toxicity and edibility information is always surfaced prominently
Offline	The app works without the internet. Cloud features enhance the experience but are not required
Design	The design is meant to ensure that every screen feels like a premium nature editorial
Privacy	Guest mode requires no account. When users sign in, their data is protected by Row level Security on Supabase

4. Target users

4.1 The explorer

Name	Emmi, 28 - Urban nature enthusiast
Context	Walks in parks, visits gardens, has a small balcony garden
Goal	Identify plants she sees and learn how to care for them
Paint point	Google searches return inconsistent results
Key feature	Camera identification, Plant details, Scan history

4.2 The survivalist

Name	Kalle, 35 - Outdoors guide and Wilderness instructor
Context	Leading hiking groups, forage walks and survival weekends
Goal	Quickly confirm whether a plant is safe to handle, eat or avoid
Paint point	Needs offline-capable identification
Key feature	Camera ID with safety focus, toxicity warning, offline mode

4.3 The plant parent

Name	Amanda, 42 - Indoor plant collector
Context	Collects houseplants, regularly buys new plants
Goal	Identify new plants and get immediate care instructions
Paint point	Care tags on plants are often wrong or missing

Key feature	Camera ID, Care tips in plant details, Scan history

5. Product goals and success metrics

5.1 Goals

Goal	Description	Priority
Identification	Users trust the identification results	Must
Instant results	Identification completes fast on a typical network	Must
Works offline	Core features and history work with no internet connection	Must
Safety-forward	Toxicity and edibility information is always visible	Must
Nice design	The digital plant editorial aesthetic should be calm and nurturing	Should
Cross- device history	Logged-in users can access their scan history	Should

5.2 Success metrics

Metric	Target	How measured
Identification success rate	> 85% of scans return a confident result	PlantNet API response scores
Time to first result	> 5 seconds	ViewModel loading state timings

Offline functionality	Core features work without internet	Manual QA in airplane mode
Crash-free rate	> 99%	Android vitals
Accessibility pass rate	100% WCAG AA for all screens	Sprint 6 audit
Unit test coverage	> 70% for ViewModel and Repository classes	JaCoCo coverage report
Academic grade	Grade 5	Submission

6. Product Scope

6.1 In scope - version 1

- Camera based plant identification
- Gallery image selection as an alternative to camera capture
- Plant organ selector(leaf, flower etc)
- Ranked species results with confidence scores
- AI-generated plant detail pages
- Local scan history stored in Room database
- Optional Supabase Authentication
- Cloud sync of scan history to Supabase
- User profile screen with scan stats
- Scan search and filter
- WCAG AA accessibility compliance

6.2 Out of scope - version 1

- iOS support
- Social features
- Community curated plant data
- AR

- Plant disease and pest detection
- Offline LLM

7. Key features

Feature	User value	Technical capability
Camera ID	Identify plants in seconds	CameraX Image capture → PlantNet API call
Safety Focus panel	Know immediately if a plant is toxic, edible or safe to touch	LLM API prompt template surfacing toxicity as top section
Organ selector	More accurate results by telling PlantNet what part of the plant you photographed	Organ parameter passed to PlantNet
Offline History	Never lose a scan even without internet or an account	Room DB write on successful identification
Cloud sync	Access scan history on any device	ScanSyncManager dual writes to Room and Supabase
AI plant details	Deep information on care, watering, soil, light and habitat	LLM API with structured prompt, cached per species to reduce API calls
Field Guide View	Browse identified plants a curated plant editorial	Editorial Compose UI pulling from local Room scan records
Guest Mode	Full core experience without creating an account	AuthState.Guest branch - local Room only, no Supabase calls

8. Assumptions and risks

8.1 Assumptions

Assumption	If wrong
PlantNet API free tier is sufficient for development and demo traffic	May need to apply rate-limits
LLM API latency is acceptable for real-time use	May need to cache responses or show progressive loading
Supabase free tier is sufficient for a class project	Minimal risk since free tier is generous for low traffic
Users will trust AI-generated toxicity information for safety descicions	Must add clear disclaimers that PlantSNap is not a substitute for expert advice
Android CameraX works reliably on target device range (API 26+)	Needs testing early to surface issues before they block later sprints

8.2 Risks

Risk	Likelihood	Impact	Mitigation
PlantNet misidentifies	Low-Medium	Critical	Show confidence score, add disclaimer on all toxicity info
LLM hallucination in plant care or toxicity data	Medium	High	Prompt-engineer for factual, conservative responses, display disclaimers
Supabase outages	Low	Low	Offline-first system means users are never blocked
CameraX permission flow rejected by users	Medium	High	Clear rationale show before permission request, graceful fallback to gallery only mode
Scope delays sprint 6 polis	High	Medium	Strict backlog prioritisation

10. Design Vision

PlantSnaps design system, The digital plant editorial. Every visual and UX interaction is guided by: Does this feel like a curated nature editorial?

10.1 Visual principles

- Organic flow
- Tonal layering instead of borders
- Typography as an editorial
- Glassmorphism on navigation - surface 70% opacity
- Shadows tinted not hard and pure black

10.2 Color Palette

Token	Hex	Role
primary	#173809	Headings, accent bars
primary-container	#2D4F1E	Gradient endpoints, secondary actions
secondary- container	#D6E7A1	Highlights
surface	#FAFAF5	Main canvas
on-surface	#1A1C19	Primary text
on-surface-variant	#43493E	Secondary

11. Timeline

Sprint	Week	Theme	Milestone
Sprint 1	Week 1	Foundation	Project skeleton, navigation, theme for design, MVVM base
Sprint 2	Week 2	Camera	Live camera preview, capture, gallery picker, organ selector
Sprint 3	Week 3	AI Identification	PlantNet API Integration, results screen, error handling
Sprint 4	Week 4	Auth and Ai Detail	Supabase Auth, profile, LLM plant details screen
Sprint 5	Week 5	Persistence and Sync	Room history, Supabase sync, search, preferences
Sprint 6	Week 6	Polish and Delivery	WCAG audit, UI tests, performance, final docs